and engineering problems. How is it possible to use a mathematics that has no connection with our everyday experience? The answer: *If the tool we have is a hammer, make the problem look like a nail.* In other words, we *change* the physical problem into a complex number form, manipulate the complex numbers, and then *change* back into a physical answer.

There are two ways that physical problems can be represented using complex numbers: a simple method of **substitution**, and a more elegant method we will call **mathematical equivalence**. Mathematical equivalence will be discussed in the next chapter on the *complex Fourier transform*. The remainder of this chapter is devoted to substitution.

Substitution takes two real physical parameters and places one in the real part of the complex number and one in the imaginary part. This allows the two values to be manipulated as a single entity, i.e., a single complex number. After the desired mathematical operations, the complex number is separated into its real and imaginary parts, which again correspond to the physical parameters we are concerned with.

A simple example will show how this works. As you recall from elementary physics, *vectors* can represent such things as: force, velocity, acceleration, etc. For example, imagine a sailboat being pushed in one direction by the wind, and in another direction by the ocean current. The resulting force on the boat is the vector sum of the two individual force vectors. This example is shown in Fig. 30-3, where two vectors, A and B, are added through the parallelogram law, resulting in C.

We can represent this problem with complex numbers by placing the east/west coordinate into the real part of a complex number, and the north/south coordinate into the imaginary part. This allows us to treat each vector as a single complex number, even though it is composed of two parts. For instance, the force of the wind, vector A, might be in the direction of 2 parts to the east and 6 parts to the north, represented as the complex number: $2 + 6j$. Likewise, the force of the ocean current, vector B, might be in the direction of 4 parts to the east and 3 parts to the south, represented as the complex number: $4 - 3j$. These two vectors can be added via Eq. 30-1, resulting in the complex number representing vector C: $6 + 3j$. Converting this back into a physical meaning, the combined force on the sailboat is in the direction of 6 parts to the north and 3 parts to the east.

Could this problem be solved without complex numbers? Of course! The complex numbers merely provide a formalized way of keeping track of the *two* components that form a *single* vector. The idea to remember is that some physical problems can be converted into a complex form by simply adding a j to one of the components. Converting back to the physical problem is nothing more than dropping the j. This is the essence of the *substitution* method.

Here's the rub. How do we know that the rules and laws that apply to complex mathematics are the same rules and laws that apply to the original